Overlay vs Hardening

Advantages and risks when applied to the data of the RFID broker

1. Your data and how it behaves

File	Written when	Must survive reboot / power cut?
config.json	When a setting is changed (web / MQTT); the Pi then reboots automatically	Yes - these are the settings
conf.d/bridge.conf	When the IP or cabinet name changes (apps.py, ipservice.py)	Yes - MQTT bridge identity and broker address
rfid2.json	After every scan	Yes - latest inventory
rfid1.json	Copied from rfid2.json at every rotation	Yes - the previous scan, used to compute add / remove
status.json	Frequently (status, session, light)	Partly (sessionId, Light); the rest is regenerated
logs/	Continuously while the logger is enabled	No - diagnostic only

2. Advantages

Overlay	Hardening
The SD card is never written, so it cannot be corrupted by power cuts. The OS and code stay clean and identical at every boot. Built into raspi-config.	All updates are kept after reboot or power cut. Each write is atomic, so files are never empty or half-written. Writes to the SD card drop sharply (logs, journal, swap go to RAM). The logger can stay enabled. No existing code is changed and no extra partition is needed.

3. What happens to each file

	With overlay	With hardening
config.json	New settings are lost at the automatic reboot that follows a change - the original problem.	New settings kept. After a power cut: complete old or complete new file.
bridge.conf	Reverts to the image version (cabinet name / broker address of the master unit).	Kept. Written atomically.
rfid1.json / rfid2.json	Revert to old data; the next scan is compared with a stale snapshot, giving false add / remove.	Kept. At most the scan in progress when power is cut is lost.
status.json	Session and light state reset.	Kept. Written atomically.
logs/	In RAM; lost at reboot; can fill RAM.	In RAM (tmpfs, 100 MB); lost at reboot; no SD writes.

4. Risks if applied to your data

Overlay

Risk	Likelihood	Consequence for your system
Settings and scan data revert at every reboot	Certain	Set changes are lost; wrong add / remove; the stated goal (keep the latest data) is not met.
ipservice.py and the cabinet name	Likely on cloned units	bridge.conf reverts to the image version; ipservice rewrites it and reboots whenever the IP letter differs from the image, which can repeat at every boot (reboot loop).
Hostname changes are lost	Certain	hostnamectl is reverted at reboot; ipservice must redo it every boot.
RAM fills with logs	Medium with the logger on	Up to about 8 MB per logger file set; system can slow down or stall.
Maintenance and remote updates	Always	Disable overlay, reboot, change, enable, reboot; hard without physical access.
Keeping some files persistent (remount of the SD card)	If attempted	Brings back write risk on the OS partition; fragile, untested; a mistake can stop booting.

Hardening

Risk	Likelihood	Consequence for your system
Power cut at the exact moment of a data write	Low (writes are short and rare once logs are in RAM)	The file stays valid; the single update in progress may be lost. SD card firmware damage is still possible on consumer cards.
sitecustomize.py behaves unexpectedly with the full application	Low; not yet tested on a Pi	Fails soft: falls back to normal file writes; removable over SSH.
fstab / logs tmpfs misconfigured	Low	Logger cannot write; boot is not blocked (nofail). Backup fstab.bak.
Logs lost at reboot	Certain	Less information to find why a cabinet restarted; mirror logs or scans to a server if needed.
Mosquitto persistence off	Certain	Queued MQTT messages do not survive a broker restart.
cmdline.txt step (optional, off by default)	Only if enabled	A mistake can prevent boot - see the separate advanced document.

5. Fit with your requirements

Requirement	Overlay	Hardening
Latest settings and scan data kept after reboot / power cut	No	Yes
Settings survive the automatic reboot after a change	No	Yes
No half-written files	Yes (nothing written)	Yes (atomic writes)
Fewer SD writes / logger usable	Yes, but logs lost and RAM can fill	Yes (logs in RAM)
No changes to existing code	Yes	Yes (new files only)
Suitable for remote cabinets	No	Yes, with the safe default

Conclusion: for this broker, whose purpose is to keep the latest RFID data and settings, **hardening is the better choice**. Overlay would protect the card but defeat the purpose of the data.

6. What neither option can guarantee

- A power cut while the SD card's own firmware is writing can damage consumer-grade cards under any setup.
- **Strongest additional protections:** an industrial or pSLC SD card, and a small UPS or supercapacitor module that triggers a clean shutdown.
- Mirror scan results to a server over MQTT so the latest data can be restored if a card fails.
- Test on one unit first (including repeated power cuts). The package has not yet been tested on a physical Raspberry Pi.